

Development Automation **& Linux Lab**

Experiment 5

Submitted By:

Ayush Singh Kushwah

SAP ID: 500095575

Enrolment No: R2142210216

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BTech CSE (DevOps)

Submitted To:

Mr. Durgesh Singh



SCHOOL OF COMPUTER SCIENCE
UNIVERSITY OF PETROLEUM & ENERGY STUDIES
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EXPERIMENT-5

Automation Scripts to save Time & Efforts

1. Set the crontab to create text file every minute in a given directory. Note that the name of the files should be different. However, you may choose any name file name.

Code:

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ cat ques1.sh
#!/bin/bash

num=$RANDOM
(touch /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/$num.txt)
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$
```

```
#Creatig a text file every minute
* * * * * /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/ques1.sh
```

Output (After 3 mins):

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ ls
13250.txt 4441.txt 9730.txt ques1.sh
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$
```

2. Write a shell script to archive all the created files (in above question 1) in the given directory. Move the archive to a given specified backup directory (different from the given directory). Insert this script as a cronjob using the crontab command, which needs to be run every 5 minutes.

Code:

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ cat ques2.sh
#!/bin/bash

tar -czf /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/logs/archieve$RANDOM.tar.gz /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/*.txt
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$
```

```
#Archiving files every 5 mins
*/5 * * * * /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/ques2.sh
```

Output:

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ ls
12775.txt 5741.txt ques1.sh ques2.sh
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ ls -l logs
total 0
-rwxrwxrwx 1 ayroid ayroid 303 Feb 27 22:40 archieve2499.tar.gz
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$
```

3. Write a shell script to automatically delete the archive files (in the backup archive) that are older than 30 minutes.

Code:

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ cat ques3.sh
#!/bin/bash

find /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/logs -mmin +30 -delete
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$

#Deleting old log archive files which are 30 mins older
* * * * * /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/ques3.sh
```

Output (After 30 mins):

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ ls
12775.txt 15390.txt 28418.txt 5741.txt logs ques1.sh ques2.sh ques3.sh
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ ls -l logs
total 0
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$
```

4. Monitor the disk usage of the given disk and alert if it is beyond the given threshold.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp5$ cat ques4.sh
#!/bin/bash

# Set the threshold for disk usage
THRESHOLD=90

# Get the disk usage as a percentage
USAGE=$(df -h | grep "/mnt/c" | awk '{print $5}' | cut -d '%' -f1)

# Check if disk usage is above the threshold
if [ $USAGE -gt $THRESHOLD ]; then
    # Send an alert
    echo "Disk usage is above $THRESHOLD% on /dev/sda1" | mail -s "Disk Alert" ayroids@gmail.com
fi

#Automating the disk usage threshold script to run every 30 mins
*/30 * * * * /mnt/c/college/upes/sem4/linux/lab/scripts/exp5/ques4.sh
```

5. Write a shell script to install the LAMP stack on Linux Machine (A-Apache, M-MySQL, P-PHP).

Code:

```
ayroid@Ayroid:/mnt/c/College/UPES/Sem4/Linux/Lab/scripts/exp5$ cat ques5.sh
#!/bin/bash

# Update the system
sudo apt update
sudo apt upgrade -y

# Install Apache
sudo apt install apache2 -y

# Install MySQL
sudo apt install mysql-server -y
sudo mysql_secure_installation

# Install PHP
sudo apt install php libapache2-mod-php php-mysql -y

# Restart Apache
sudo systemctl restart apache2
```